Casey Barton

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PROFILE

Applied-AI systems engineer based in Honolulu. I design, build, deploy, and scale production AI/ML capabilities — bridging AI research, data science, software engineering, and product delivery. My systems run retrieval-grounded generation over federated knowledge (Mem0, Supermemory, Supabase pgvector, Pinecone, Neo4j behind one MCP interface), recommendation-grade personalization loops, and the eval discipline that keeps production AI honest.

Three years running an evidence-grade agent control plane: typed state transitions, approval gates, audit receipts, recoverable runs — the governance layer production AI/ML needs to ship. A 1,395-repository estate with **65 MCP connectors**, a **989-test** suite, **62/62** verification checks green. Strategic and hands-on: I set platform direction and ship the systems myself.

CORE COMPETENCIES

Engineering: Python · TypeScript · Node.js · SQL · Bash · REST APIs · Supabase · Postgres · Vercel · GitHub Actions (CI/CD)

AI / ML: production AI/ML design, build, deploy & scale · LLM retrieval & knowledge graphs · recommendations & personalization · LLM evaluation harnesses · prompt & context engineering · MCP protocol · structured outputs

Delivery: research-to-production bridging · verification gates · audit receipts · observability · fail-closed recovery · clear human-facing documentation

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Design, build, deploy, and scale production AI capabilities: retrieval-grounded generation with provenance (unified-memory-mcp, mcp-iceberg-mansion) — federated knowledge graphs over Mem0, Supermemory, Supabase pgvector, Pinecone, and Neo4j; parallel fan-out, content-hash dedup.
- Evaluation-first delivery (job-app-helix): **989-test** agent control plane; evals with explicit release thresholds, gold sets and LLM-as-judge — the harness that decides whether an AI capability ships, bridging research prototypes and production delivery.
- Multi-agent orchestration (anthropic-agent-coordinator): deterministic DAG scheduler, **62/62** tests green; tool calls proposed for review, never executed blind — personalization and recommendation loops with human-in-the-loop control.
- Production reliability (apex-control-plane): SHA-256 transport envelopes, idempotency keys, circuit breakers, dead-letter capture, fail-closed startup; failure analysis with root-cause quarantine instead of silent retries.
- Operate a **1,395-repository** applied-AI estate — **65 MCP connectors** integrated with claim tracking and idempotent automation; reusable agents, skills, connectors, and playbooks as the compounding output; **20+ production repositories** maintained under test and CI.

Chief Inspector 2024

Diamond Head Home Inspections — Honolulu, HI

- Led residential inspection operations: field assessments, defect documentation, and client reporting — direct customer-facing technical work with non-technical stakeholders.

EDUCATION

University of Hawaii at Manoa 2016

Bachelor of Science

AWS Cloud Institute 2025

Cloud Computing program (brief attendance)

Colorado State University

Undergraduate coursework

SELECTED PROOF

Job-App Helix — attributable campaign engine

- Composes assessments into campaign-level **GO / NO-GO** with immutable evidence models, stable finding codes, and fail-closed decisions — review gates with real veto power.
- Recovery restricted to contingency evidence declared before reassessment — the engine never silently tunes values until they pass.

GlacierEQ estate — agent systems portfolio

- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate — the governance layer production AI needs.
- LLM-as-judge eval harnesses over retrieval, tool use, and multi-step agent behavior — measured gains, not asserted ones.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

AI / ML: production AI/ML systems · LLM retrieval & knowledge graphs · recommendations · LLM evaluation · prompt & context engineering

Data & Infra: Supabase · Postgres · Vercel · GitHub Actions · REST APIs · MCP tool integration patterns

Practice: evals · review gates · observability · fail-closed recovery · research-to-production delivery

Evidence boundary: portfolio repositories are independent work. Domain-inspired demonstrations are not claims of company affiliation, production deployment at Pearson, ed-tech domain production depth, or certification.